

Verbatim Conversation Transcript: Network DDoS and Spam Email

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Transcript Overview

This document provides a word-for-word reproduction of the study session regarding the provided lecture notes on DDoS and Spam. Every prompt and response is included in its original form.

User

[File Uploaded: 12_3_DDoS_Spam.pdf] *(The user initiated the session by providing the lecture slides.)*

Gemini

Based on the provided lecture notes, Network Denial of Service (DoS) attacks aim to disrupt large sites using minimal computing effort through **amplification**. These attacks can occur at various layers, including Link, TCP/UDP, Application, and Payment layers.

Types of Network DoS Attacks

- **DoS Bug:** A design flaw that allows a single machine to disrupt a service.
- **DoS Flood:** Utilizing a botnet to generate a massive flood of requests.
- **Amplification Attacks:**
 - **Smurf Attack:** An attacker sends an ICMP Echo Request (ping) to a broadcast address, spoofing the victim's IP as the source. Every host on that network then replies to the victim, creating a flood.
 - **DNS Amplification:** An attacker sends a small DNS query (e.g., 60 bytes) with a spoofed source IP to an open resolver, which responds with a much larger record (e.g., 3000 bytes) to the victim.
 - **Reflector Attacks:** These use network components (like DNS resolvers or web servers) to respond to spoofed packets, hiding the attacker's botnet and making traceback difficult.

TCP SYN Floods A common DoS method involves the TCP handshake. An attacker sends **SYN packets** with random source IP addresses, filling the server's **backlog queue** and preventing new legitimate connections.

- **Low Rate SYN Flood:** Because some operating systems have small backlog queues (e.g., WinNT 4.0 only allows 6) and long timeouts (3 minutes), an attacker might only need to send 128 packets every 3 minutes to keep a service down.
- **Massive Flood:** Large-scale Distributed DoS (DDoS) can use 20,000+ bots to generate gigabits of traffic, saturating network uplinks.

Defenses against SYN Floods

- **Syncookies:** A "correct" solution that removes the need for the server to store state until the handshake is verified.
- **Client Puzzles:** Forcing the client to solve a moderately hard mathematical problem (like a SHA-1 hash challenge) to slow down the attacker.
- **CAPTCHAs:** Verifying the connection is from a human, often used for application-layer attacks.

Spam Email and Verification Spam is often sent through **Open Relays**, which forward mail and can be used to hide the original source IP. To combat spam and email spoofing, two primary verification methods are used:

- **SPF (Sender Policy Framework):** A DNS record that lists the authorized IP addresses allowed to send mail for a specific domain.
- **DKIM (Domain Keys Identified Mail):** The sender's Mail Transfer Agent (MTA) cryptographically signs the email. The receiver verifies this signature using a public key found in the sender's DNS records.

Would you like more detail on the specific mathematical implementation of client puzzles or how DKIM signatures are structured?

User

slide 3

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Slide 3 of the lecture notes details the **Smurf amplification DoS attack**, which uses the ICMP protocol to overwhelm a victim.

How the Attack Works

- **The Request:** The attacker sends an **ICMP Echo Request** (ping).
- **Spoofing:** The source IP of this request is spoofed to be the **DoS Target's IP**, while the destination is set to a **network broadcast address**.
- **The Amplification:** Because it is sent to a broadcast address, the gateway delivers the request to every host on that network.
- **The Flood:** Every host on the target network then generates an **ICMP Echo Reply** directed at the victim (the DoS Target).

Prevention

- The primary defense against this specific attack is to configure networks to **reject external packets** sent to a broadcast address.
- Would you like to see how the **DNS Amplification** attack on the next slide compares to this ICMP-based method?

User

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Slide 4 of the lecture notes provides a **modern-day example** of a DoS attack using **DNS Amplification**, which achieved a $50\times$ **amplification** factor.

Mechanics of the DNS Amplification Attack

- **The Query:** The attacker sends a **DNS Query** (typically 60 bytes) to a DNS server.
- **Spoofing:** The source IP of the query is spoofed to match the **DoS Target's IP**.
- **The Response:** The DNS server sends an **EDNS Response** (which can be as large as 3000 bytes) to the victim.
- **Scale:** In 2006, there were approximately **580,000 open resolvers** on the Internet that could be exploited this way.

— **Comparison: Smurf vs. DNS Amplification**

Feature	Smurf Attack (Slide 3)	DNS Amplification (Slide 4)
Protocol	ICMP (Ping)	DNS (UDP)
Strategy	Broadcast to a local network	Querying open resolvers
Key Weakness	Relying on broadcast addresses	Large response-to-query ratio

Would you like to move on to **Slide 5**, which explains how **TCP SYN Floods** work at the transport layer?

User

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Slide 5 illustrates a **Low Rate TCP SYN Flood**, which is classified as a “DoS bug” because it exploits a specific design flaw in how servers handle connection requests.

The Attack Mechanism

- **The Actor:** A single machine initiates the attack.
- **The Method:** The attacker sends a series of **SYN packets** (SYN_{C1} through SYN_{C5} in the diagram) to a server.
- **IP Spoofing:** Each of these packets uses a **random source IP address**.
- **Resource Exhaustion:** These requests fill up the server's **backlog queue**.

- **The Result:** Once the queue is full, the server can no longer accept any further legitimate connections.

Why it works The server allocates resources (state) for every incoming SYN packet, waiting for a response that never comes because the source IPs are fake.

— Would you like to see **Slide 6**, which shows exactly how few packets it takes to crash specific operating systems like Linux or Windows NT?

User

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Slide 6 provides specific data from 1996 to illustrate why a **Low rate SYN flood** is so effective against common operating systems.

Backlog Queue Limits The efficiency of this attack depends on the maximum number of half-open connections an OS can track at once:

Operating System	Backlog Queue Size
Linux 1.2.x	10
FreeBSD 2.1.5	128
WinNT 4.0	6

The “Low Rate” Math

- **Timeout:** The typical backlog timeout is **3 minutes**.
 - **Required Effort:** To keep a FreeBSD server (the strongest of the three) completely unreachable, an attacker only needs to send **128 SYN packets every 3 minutes**.
 - **Impact:** This tiny amount of traffic—less than one packet per second—is enough to exhaust the server’s resources and block all legitimate users.
- Would you like to see **Slide 7**, which describes a real-world 2003 example of this attack used by the **MS Blaster worm**?

User

yes

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Slide 7 provides a real-world case study of these principles in action: the **MS Blaster worm** from 2003.

The MS Blaster Attack

- **Target:** The worm was programmed to launch a SYN flood against **windowsupdate.com** on August 16th.
- **Intensity:** Infected machines sent **50 SYN packets every second** to port 80.
- **Packet Details:** Each packet was small (**40 bytes**) to maximize efficiency.
- **Obfuscation:** It used **spoofed source IP addresses** in the format a.b.X.Y, where X and Y were randomized to hide the identity of the infected “zombies”.

Microsoft’s Countermeasures Instead of trying to stop the flood directly, Microsoft used redirection and third-party infrastructure:

- **Domain Change:** They moved the service to a new name: **windowsupdate.microsoft.com**.
- **Content Delivery Network (CDN):** They used **Akamai** to deliver the update files, distributing the load across a massive global network that was much harder to overwhelm than a single site.

— Would you like to see **Slide 8**, which explains the “correct” technical solution for servers to defend against these floods?

User

akamai??

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Slide 7 explains how Microsoft used **Akamai** as a primary defense strategy against the **MS Blaster worm** in 2003.

Why Akamai? Instead of trying to absorb the SYN flood on their own servers, Microsoft offloaded the traffic to Akamai's infrastructure:

- **Content Delivery Network (CDN):** Akamai operates a massive, globally distributed network of servers.
- **Infrastructure Shifting:** By moving the Windows Update files to Akamai, the attack was no longer hitting a single central target.
- **Domain Redirection:** Microsoft changed the service name to `windowsupdate.microsoft.com` to help manage the transition and bypass the hardcoded target in the worm's code (`windowsupdate.com`).

The Result The MS Blaster worm was programmed to flood a specific domain at a specific time. By using Akamai, the "load" of the update requests (and the attack traffic) was spread across thousands of edge servers, making it nearly impossible for the worm to "take out" the service.

— Would you like to move to **Slide 8** to see the **Syncookies** solution, which is the "correct" technical fix for the SYN flood bug?

User

yes

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Slide 8 outlines the technical defenses against the low-rate SYN floods discussed in the previous slides.

Non-Solutions Simply trying to "out-resource" the attack usually fails:

- **Increasing backlog queue size:** The attacker can simply send more packets to fill the larger queue.
- **Decreasing timeout:** Setting the timeout too low might drop legitimate connections from users with slow internet speeds.

The Correct Solution: Syncookies The most effective defense when under attack is using **Syncookies**.

- **Remove State:** Normally, a server allocates memory (state) as soon as it receives a SYN packet. Syncookies allow the server to **remove state** entirely during the initial handshake.
- **How it works:** Instead of storing the connection request in a queue, the server sends back a SYN-ACK where the **Sequence Number** is a specially crafted "cookie" (a hash of the connection details).
- **Validation:** The server only allocates memory if the client responds with a valid ACK that matches the cookie.
- **Performance:** This method has a **small performance overhead** but effectively prevents the backlog queue from filling up with fake requests.

— Would you like to move to **Slide 9**, which discusses what happens when the attack scales up to a **Massive DDoS flood**?

User

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Slide 9 shifts the focus from "low rate" bugs to **Massive SYN Floods**, which are a form of Distributed Denial of Service (DDoS).

The Scale of the Attack

- **Bot Armies:** Attackers command a "bot army" to flood a specific target simultaneously.
- **Volume:** As far back as 2003 (e.g., the BetCris.com attack), 20,000 bots could generate $2Gb/sec$ of SYN packets.
- **Infrastructure Impact:** This level of traffic saturates the target's network uplink or their network router.

The Identification Problem

- **Random Source IPs:** Because attackers use random source IP addresses, the malicious SYN packets look identical to legitimate ones.
- **Defense Difficulty:** This makes it extremely hard for a server to distinguish between a real user trying to connect and a bot packet.

— Would you like to see **Slide 10**, which explains even stronger attacks that can bypass simple SYN flood protections?

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Slide 10 describes **TCP Connection Floods**, which are more sophisticated than simple SYN floods because they involve completing the full three-way handshake.

How the Attack Works

- **The Command:** An attacker commands a bot army to interact with a website.
- **Full Handshake:** Unlike a SYN flood, the bots **complete the TCP connection**.
- **Application Request:** Once connected, they send a short **HTTP HEAD request** and repeat the process.

Why It Is Stronger

- **Bypasses Defense:** This method will **bypass SYN flood protection proxies** because the connection appears “legitimate” since it was fully established.
- **The Trade-off for Attackers:** To complete the handshake, the attacker **can no longer use random source IP addresses**.
- **Vulnerability:** Because the real IP of the bot is used, it **reveals the location of the bot zombies**.
- **Defense Strategy:** Once the IPs are known, a proxy or firewall can **block or rate-limit** those specific bots.

— Would you like to see **Slide 11**, which moves the discussion to **DNS DoS attacks** and the “collateral damage” they cause?

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Slide 11 focuses on **DNS DoS Attacks**, using the 2006 attack on Blue Security as a primary example.

Vulnerabilities in DNS

- **Protocol:** DNS typically runs on **UDP port 53**.
- **Targeting:** An attacker identifies the DNS entry for a victim (e.g., `victim.com`) which is hosted at a specific provider (e.g., `victim_isp.com`).

The DDoS Attack Method

- **Flooding:** The botnet floods the ISP’s DNS server with requests for the victim’s domain.
- **Obfuscation:** Like SYN floods, these UDP packets use **random source IP addresses**.

The Impact: Collateral Damage

- **Service Failure:** The volume of traffic can take out the **entire DNS server**, not just the victim’s record.
- **Case Study:** When Blue Security was attacked, their DNS was hosted at Tucows; the resulting DDoS took out the Tucows servers, affecting “many many sites” besides the original target.

— Would you like to move to **Slide 12**, which discusses DoS at the **Application layer**, specifically involving **SSL/TLS** handshakes?

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Slide 12 covers **DoS at higher layers**, specifically focusing on the **SSL/TLS handshake** and how it can be used to overwhelm a server.

SSL/TLS Handshake Vulnerability The core issue is an **asymmetry in computational cost** between the client and the server during the RSA key exchange.

- **The Process:** The client sends a “Client Hello,” the server responds with a “Server Hello” and its public key, and then the client sends an encrypted key.
- **The Math:** RSA encryption (done by the client) is roughly $10\times$ **faster** than RSA decryption (done by the server).
- **The Impact:** Because the server has to do much more “work” than the client, a **single machine** can potentially bring down **ten web servers** by forcing them to perform constant decryptions.

Application Layer DoS This same principle of “easy for client, hard for server” applies to other application-level requests:

- **Large File Requests:** An attacker can request a very large file, such as a **large PDF**, with a tiny HTTP request.
- **Resource Exhaustion:** This requires almost no effort from the client but forces the server to do significant work in fetching and transmitting the data.

Would you like to see **Slide 13**, which introduces **Client Puzzles** as a way to force attackers to do more work?

User

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Slide 13 introduces **Client Puzzles** as a defensive strategy designed to neutralize the “easy for client, hard for server” asymmetry.

The Core Concept The goal is to force the client to prove they have expended a certain amount of computational effort before the server commits its own resources.

- **The Challenge:** The server gives the client a challenge C .
- **The Puzzle:** The client must find a value X such that the least significant n bits of a SHA-1 hash ($SHA-1(C||X)$) are all zeros (0^n).
- **The Effort:** This is a “moderately hard” problem that takes an expected 2^n time to solve. For $n = 16$ this takes roughly **0.3 seconds** on a 1GHz machine.
- **The Benefit:** While the puzzle is hard to solve, it is **extremely easy for the server to verify** the solution.

Implementation During an Attack

- **Dynamic Requirement:** Under normal conditions, the server does not require puzzles.
- **Under Attack:** When a DoS attack is detected, the server requires everyone to submit a valid puzzle solution with their requests.
- **Impact on Attackers:** This drastically slows down an attacker trying to send thousands of requests per second, as their hardware becomes a bottleneck.

— Would you like to see **Slide 14**, which discusses how **CAPTCHAs** serve a similar purpose but for human verification?

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Slide 14 introduces **CAPTCHAs** as a method to defend against application-layer DDoS attacks by verifying the presence of a human user.

The Core Concept

- **Human Verification:** The fundamental idea is to ensure that the connection is being initiated by a person rather than an automated bot.
- **Visual Challenges:** The slide displays examples of distorted text such as “STRAIGHT,” “EARLY,” and “BEAUTIFUL” that users must identify.

Implementation During an Attack

- **Selective Triggering:** CAPTCHAs are typically generated and required only during an active attack.
- **Request Processing:** The server will only process an application request if the user provides a valid solution to the CAPTCHA.
- **Rate Limiting:** To prevent bots from simply solving one and continuing the attack, the system presents **one CAPTCHA per source IP address**.

— Would you like to move to **Slide 15**, which begins the discussion on **Source Identification** and the ultimate goal of blocking attacks at their origin?

User

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Slide 15 introduces the concept of **Source Identification**, moving from immediate mitigation to the long-term goal of stopping an attack at its origin.

Core Objectives

- **Identify Packet Source:** The primary technical challenge is determining the actual origin of attack traffic, especially when IP spoofing is involved.
- **Block at the Source:** The ultimate goal of identification is to enable defenders to block the attack as close to the source as possible, preventing it from ever reaching the target's network.

This slide serves as a transition into the more complex **Reflector Attacks** shown on **Slide 16**, which make this identification process significantly harder. Would you like to see how reflectors are used to hide an attacker's identity?

User

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Slide 16 explains **Reflector Attacks**, a technique used by attackers to hide their identity and amplify their impact by using legitimate network components as intermediaries.

What is a Reflector?

- A **reflector** is any common network component that is configured to respond to incoming packets.
- The attacker sends a packet to the reflector but **spoofs the source IP address** to be that of the victim.
- The reflector then sends its response directly to the **victim**, rather than the actual attacker.

Common Examples of Reflectors

- **DNS Resolvers:** An attacker sends a UDP port 53 request with the victim's source IP; the resolver then sends the DNS response to the victim.
- **Web Servers:** An attacker sends a TCP SYN to port 80 with the victim's source IP; the web server then sends a **TCP SYN ACK** packet to the victim.
- **Gnutella Servers:** These peer-to-peer servers can also be manipulated to act as reflectors.

— Would you like to see **Slide 17**, which provides a visual map of how a single "Master" controls a massive army of these reflectors to kill traceback efforts?

User

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Slide 17 provides a visual map of a complex **Reflector DDoS Attack**, showing how an attacker can use multiple layers of infrastructure to stay hidden while overwhelming a victim.

The Hierarchy of the Attack

- **Single Master:** A single entity controls the entire operation, sending control traffic to direct the next layer.
- **Slave Bots:** The Master directs several “Slaves” (infected bot machines) to send requests to reflectors.
- **Zillions of Reflectors:** The Slaves send requests with the **victim’s spoofed source IP** to a massive number of reflectors (DNS, Web, or Gnutella servers).
- **Unsolicited Traffic:** The reflectors then send a massive stream of “replies” to the victim. Since these reflectors are legitimate servers, the traffic they send is **non-spoofed**, making it look like legitimate (though unsolicited) data.

Why This is Effective

- **Hides the Bots:** Because the victim only sees traffic coming from the reflectors, the actual bot army (the Slaves) remains hidden.
- **Defeats Defenses:** This architecture “kills” traditional **traceback** and **pushback** methods because the trail of breadcrumbs leads back to innocent reflectors rather than the attacker’s bots.

— This concludes the section on Denial of Service. Would you like to move on to **Slide 19**, which starts the next topic: **How email and Spam work?**

User

19

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Slide 19 breaks down the basics of how email is sent using the **Simple Mail Transfer Protocol (SMTP)**, which has been the standard since RFC 821 in 1982.

Common SMTP Commands The delivery process involves a specific sequence of commands between the sender and the receiver:

- **MAIL FROM: <reverse-path>:** Identifies the sender of the message.
- **RCPT TO: <forward-path>:** Identifies the recipient. This command is repeated for each additional recipient.
 - If the recipient is unknown, the server responds with a “550 Failure reply”.
- **DATA:** This command initiates the transfer of the actual email headers and the body content.
- **VRIFY username:** Used to verify if a specific user exists on the server.
 - It returns a **250** if the user exists or a **550** if they do not.
 - **Note:** This is often disabled to prevent spammers from harvesting valid email addresses.

— Would you like to see **Slide 20**, which explains how **Blacklists** are used to stop spammers who abuse these commands?

User

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Slide 20 introduces **Spam Blacklists**, a primary defense mechanism used to block unwanted emails during the initial connection phase.

Realtime Blackhole Lists (RBL) RBLs are databases that track and list servers or ISPs known for generating large volumes of spam. Common examples include:

- spamhaus.org
- spamcop.net

Effectiveness Data from spamhaus.org highlights how these lists perform:

- **Connection Level:** RBLs can stop approximately **15-25%** of incoming spam at the moment the SMTP connection is attempted.

- **Content Level:** Over **90%** of spam can be stopped when combined with message body URI checks.

The Spammer's Counter-Move Because these lists are so effective, the primary goal for a spammer is to **evade blacklists** by hiding or frequently changing their source IP address.

— Would you like to move to **Slide 21**, which explains how spammers use **Open Relays** to achieve this goal of hiding their identity?

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Slide 21 explains how spammers use **Open Relays** to bypass the blacklists mentioned on the previous slide.

How an SMTP Relay Works An SMTP relay is a server that accepts email from one source and forwards it to its final destination. Spammers exploit this using the following steps:

1. **Connection:** A bulk email tool connects to the relay via **SMTP port 25**.
2. **Recipient List:** The spammer sends a list of targets using the `RCPT TO` command.
3. **Single Transmission:** The spammer sends the email body only **once** for all recipients, saving their own bandwidth.
4. **Delivery:** The relay server then handles the actual delivery to the final recipients.

Honest vs. Hacked Relays

- **Honest Relay:** It adds a **Received header** to the email, which reveals the original source IP address of the spammer.
- **Hacked Relay:** It is configured or modified so that it **does not** include the source IP, effectively hiding the spammer's identity from blacklists.

— Would you like to move to **Slide 22**, which covers **SPF**, the first of two major methods for verifying that a sender is who they claim to be?

User

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Slide 22 introduces **Sender Policy Framework (SPF)**, a DNS-based mechanism designed to prevent email spoofing.

The Problem: Spoofing Spammers often send emails claiming to be from a reputable domain like `hotmail.com`. This results in "bounce messages" (delivery failure notifications) flooding the legitimate domain's system.

How SPF Works SPF allows a domain owner to specify exactly which mail servers are authorized to send emails on their behalf.

- **DNS Record:** The domain owner (e.g., `hotmail.com`) publishes a special **TXT record** in their DNS settings.
- **Authorization List:** This record contains a list of authorized IP addresses (e.g., `64.4.33.7`).
- **Verification:** When an email arrives, the recipient's mail server checks the **"MAIL FROM"** address.
- **The Check:** The recipient server queries the DNS of the claimed sender domain to see if the **SenderIP** is in the authorized list.
- **Hard Fail:** A record ending in `-all` (like the example `v=spf1 a:mailers.hotmail.com -all`) instructs the receiver to reject any mail that doesn't come from the listed IPs.

— Would you like to move to **Slide 23** to see **DKIM**, which uses cryptographic signatures for even stronger verification?

User

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Slide 23 introduces **Sender verification II: DKIM** (Domain Keys Identified Mail), which provides a more robust way to prevent email spoofing than SPF.

The Core Concept of DKIM While SPF checks the sender's IP address, DKIM uses **cryptographic signatures** to verify the email's origin and integrity.

- **Signing:** The sender's Mail Transfer Agent (MTA) signs the email, including the body and specific header fields.
- **Verification:** The receiver's Mail User Agent (MUA) checks this signature.
- **Rejection:** If the signature is invalid, the email is rejected.
- **Public Key Management:** The sender's public key is stored in a DNS TXT record, often under a subdomain like `_domainkey.hotmail.com`.

Why use DKIM? It is considered harder to spoof than SPF because it relies on a private key held only by the legitimate sender's server.

— Would you like to see **Slide 24**, which provides a technical breakdown of what a **DKIM-Signature header** actually looks like?

User

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Slide 24 provides a technical breakdown of a **DKIM-Signature header**, illustrating how the cryptographic data is structured within an email.

Breakdown of the DKIM Header The header contains several key-value pairs (tags) that tell the receiving server how to verify the message:

- `a=rsa-sha1`: Specifies the algorithm used for the signature.
- `d=hotmail.com`: The **domain** claiming responsibility for the message.
- `s=may2006`: The **selector**, which helps the receiver find the correct public key in the DNS.
- `h=from:to:subject:date`: Lists the specific **header** fields that were included in the cryptographic hash.
- `b=...`: The actual **cryptographic signature** (the hash of the headers and body, encrypted with the sender's private key).
- `t` and `x`: Represent the **timestamp** and **expiration** time of the signature.

Verification Process To verify this specific example, the recipient's mail software (MUA) performs a DNS query:

1. It combines the selector and the domain to form a specific address: `may2006._domainkey.hotmail.com`.
2. It requests the **DNS TXT record** for that address to retrieve the public key.
3. It uses that public key to decrypt the signature and ensure the email hasn't been tampered with.

This concludes the provided lecture slides on DDoS and Spam. Is there a specific attack or defense mechanism you'd like to dive deeper into?